Here's a comprehensive guide for handling and storing your CsI(Tl) crystal:

CsI(Tl) Crystal Handling & Storage Guide

Key Properties to Remember

CsI is slightly hygroscopic. Contact with water and high humidity should be avoided.

CsI is relatively soft and plastic and does not cleave. Because it has no cleavage plane, it is quite rugged.

Good news: CsI(Tl) is much more forgiving than NaI(Tl) — it won't dissolve if you briefly expose it to air like NaI will.

Unpacking New Crystals

CAUTION: DO NOT OPEN PACKAGE UNTIL DETECTOR REACHES ROOM TEMPERATURE!

If the detector package comes into the laboratory from a truck or warehouse where the temperature differs by 5°C (10°F) or more from that of the laboratory, allow the package to reach room temperature before opening. This will prevent fracturing the crystal from thermal shock. A good practice is to leave the package in the laboratory overnight before opening.

This is especially relevant for you in Japan — packages from China may have experienced significant temperature changes during shipping.

Temperature & Thermal Shock

NEVER STORE THE DETECTOR NEAR A HEATING ELEMENT, SUN-WARMED SURFACE, RADIATOR OR AIR CONDITIONER!

Detectors are intended for use in a normal laboratory environment. They will operate reliably between 4°C and 43°C (40°F and 110°F), provided the rate of temperature change does not exceed 8°C (15°F) per hour.

For your floating house environment: Be mindful of temperature swings — gradual changes are fine, rapid ones can crack the crystal.

Humidity & Moisture Protection

CsI(pure), CsI(Tl), and BaF_2 crystal surfaces are easily damaged by drops of moisture or excessive condensation.

Practical steps:

1. **Never touch the polished optical faces** with bare fingers — oils and moisture will damage the surface
2. **Use clean, lint-free gloves** (nitrile or cotton)
3. **Avoid breathing directly on the crystal**
4. **Work in low-humidity conditions** when possible

Good news from research: The findings indicate that there was no significant impact on performance in terms of resolution after almost three years under adverse conditions of humidity and temperature. This suggests that the employed wrapping for light collection effectively serves as an excellent barrier against humidity.

So once you wrap your crystal properly, it's well protected.

UV Light Exposure

Ultraviolet radiation in sunlight or fluorescent lighting can produce discoloration and phosphorescence in scintillation crystals. The coloration produced by UV radiation appears in the bulk of the crystal rather than the surface and is most noticeable in large crystals. For this reason, open window detectors or unpackaged crystals should be stored in darkness when not in use.

Practical steps:

- Store unwrapped crystals in a dark container
- Don't leave crystals sitting on a windowsill
- Once wrapped in PTFE/Teflon and housed, UV is blocked

Storage Recommendations

Do	Don't
Store in original packaging until ready to use	Expose to direct sunlight
Keep in a dry location (use silica gel packets)	Store near heat sources
Store in darkness	Touch optical surfaces
Wrap in PTFE tape when assembling	Expose to water or high humidity
Allow temperature equalization before opening	Subject to rapid temperature changes

Long-term storage:

- Original sealed packaging is best
- If opened, store in a sealed container with desiccant (silica gel)
- Keep at stable room temperature
- Wrap in clean tissue or PTFE tape

Handling During Assembly

1. **Wear nitrile gloves** — cotton gloves can leave fibers
2. **Handle by the edges**, never the optical coupling face
3. **Work quickly** but carefully when the crystal is exposed
4. **Prepare all materials beforehand** (PTFE tape, optical grease, housing)
5. **If you drop it** — CsI(Tl) is quite rugged and may survive, but inspect for chips or cracks

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Signs of Damage

In general, deterioration of energy resolution or of the absolute efficiency is an indicator of detector degradation.

Visual signs to watch for:

- Cloudiness or hazing on surfaces (moisture damage)
- Yellowing or discoloration (UV damage or hydration)
- Chips, cracks, or surface pitting
- Sticky or tacky surface feel

If surface gets damaged:

- Minor surface contamination can sometimes be carefully cleaned with dry isopropanol and lint-free wipes
- Severe damage may require re-polishing (usually means replacing for small crystals)

Quick Checklist for Your Build

- [ ] Let crystal package reach room temperature overnight before opening
- [ ] Prepare workspace: PTFE tape, optical grease, gloves, housing ready
- [ ] Work in low-humidity environment
- [ ] Wear nitrile gloves
- [ ] Inspect crystal for damage before assembly
- [ ] Wrap quickly in PTFE tape (all sides except SiPM coupling face)
- [ ] Apply optical grease to coupling face
- [ ] Mount SiPM and seal in light-tight housing
- [ ] Store any spare crystals in sealed container with desiccant

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The bottom line: CsI(Tl) is one of the more forgiving scintillator crystals to work with. It's not nearly as sensitive as NaI(Tl). Just avoid water, rapid temperature changes, and prolonged UV exposure, and you'll be fine.